

Graph Theory Assignment 2 Report

Group 5

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Group Homework

Adapted from MIT Urban Operations Research, with modifications

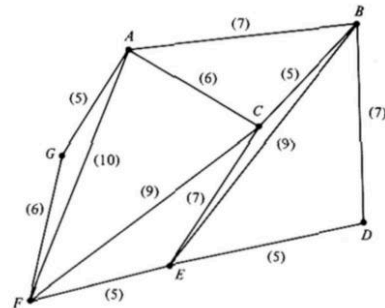
You are given this graph network on the right side

You must:

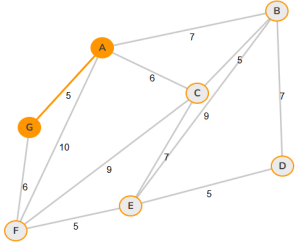
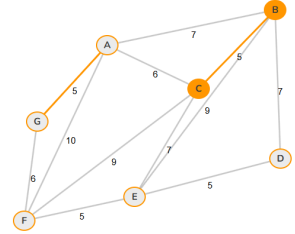
1. Implement Prim's algorithm to find MST
2. Implement Kruskal's algorithm to find MST
3. Independently find another algorithm than any explained on this slide, then implement it to find MST
4. Simulate if any node/edge failed (cannot be traversed), report how your algorithm adapt

You must submit

- A. GitHub repository containing the implementation
- B. A PDF report of 1-4, committed in the repository



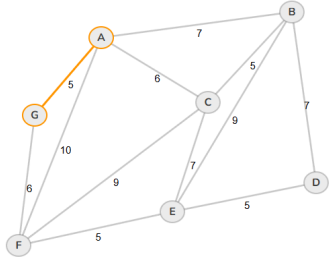
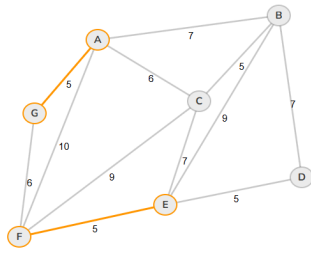
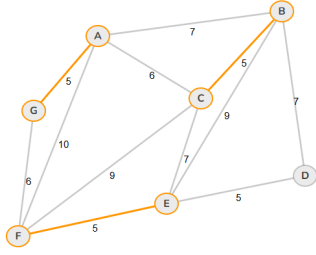
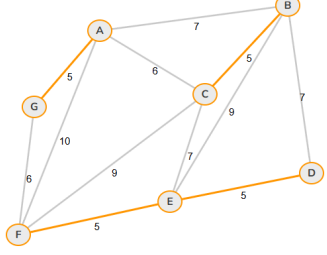
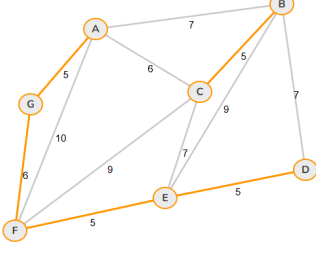
Boruvka

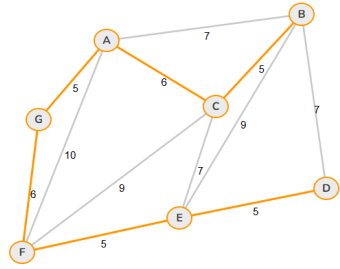
Step	U	Edg(u, v)	V - U	Graph
0	{ }	-	{ A ,B ,C ,D ,E ,F ,G }	
1	{A, G}	(A - G) = 5	{B, C, D, E, F}	
2	{A, G} {B, C}	(B - C) = 5	{D, E, F}	

3	{A, G}, {B, C}, {D, E}	(D - E) = 5	{F}	
4	{A, G}, {B, C}, {D, E, F}	(E - F) = 5	{}	
5	{A, B, C, G}, {D, E, F}	(A - C) = 6	{}	
6	{A, B, C, D, E, F, G}	(F - G) = 6	{}	

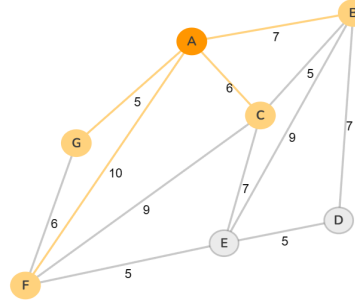
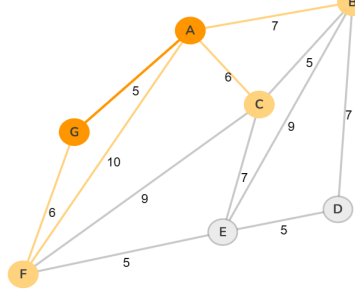
Kruskal

Step	U	Edg(u, v)	V - U	Graph
0	{}	-	{ A ,B ,C ,D ,E ,F ,G }	-

1	{ A , G }	$(A - G) = 5$	{ B , C , D , E , F }	
2	{ A , G } , { B , C }	$(B - C) = 5$	{ D , E , F }	
3	{ A , G } , { B , C } , { D , E }	$(D - E) = 5$	{ F }	
4	{ A , G } , { B , C } , { D , E , F }	$(E - F) = 5$	{ }	
5	{ A , B , C , G } , { D , E , F }	$(A - C) = 6$	{ }	

6	{ A , B , C , D , E , F , G }	(F - G) = 6	{ }	
7	{ A , B , C , D , E , F , G }	(A - B) = 7 cycle(B - D) = 7 cycle(C - E) = 7 cycle(B - E) = 9 cycle(C - F) = 9 cycle(A - F) = 10 cycle	{ }	

Prim

Step	U	Edg(u, v)	V - U	Graph
0	{ A }	-	{ B , C , D , E , F , G }	
1	{ A , G }	(A - G) = 5	{ B , C , D , E , F }	

2	{ A, G, C }	$(A - C) = 6$	{ B, D, E, F }	
3	{ A, G, C, B }	$(C - B) = 5$	{ D, E, F }	
4	{ A, G, C, B, F }	$(G - F) = 6$	{ D, E }	
5	{ A, G, C, B, F, E }	$(F - E) = 5$	{ D }	

6	{ A, G, C, B, F, E, D }	(E - D) = 5	{ }	
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Failure Simulation:

- Boruvka
- **Node failure: F goes down**
F is a high-degree node (connected to A, C, E and G). Removing it also removes all 4 of its incident edges. The algorithm still finds a valid MST over the remaining 6 nodes, but D-E-F's cheap E-F(5) link and G's F-G(6) link both disappear, forcing G to rely solely on A-G(5) (still cheap) and pushing the overall structure to rely more on A-C(6) and B-D(7).

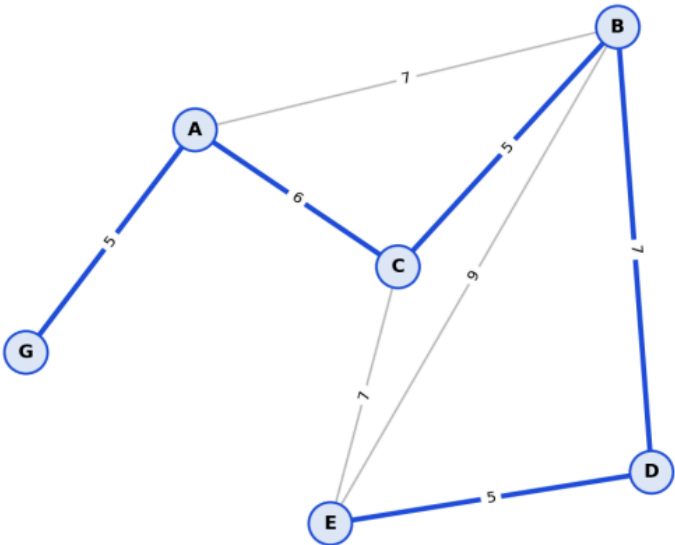
Round	Components before	Cheapest edge per component	Components after
1	{A}, {B}, {C}, {D}, {E}, {G}	(A-G)=5, (B-C)=5, (D-E)=5	{A,G}, {B,C}, {D,E}
2	{A,G}, {B,C}, {D,E}	(A-C)=6, (B-D)=7	{A,B,C,D,E,G}

New MST edges: (A-G)=5, (B-C)=5, (D-E)=5, (A-C)=6, (B-D)=7

New total weight: 28 (over 6 nodes instead of 7)

Graph still fully connected: True

Scenario 2: Node F Fails (Total Weight = 28)



- **Edge failure: A–G (weight 5) goes down**

Edge A-G was part of the original MST. With it removed, G can now only reach the rest of the graph through F-G (weight 6). Boruvka's algorithm adapts automatically: in round 1 it simply cannot find a cheaper option for G's component than F-G, so it substitutes that edge instead.

Round	Components before	Cheapest edge per component	Components after
1	{A}, {B}, {C}, {D}, {E}, {F}, {G}	(A-C)=6, (B-C)=5, (E-F)=5, (D-E)=5, (F-G)=6	{A,B,C}, {D,E,F,G}
2	{A,B,C}, {D,E,F,G}	(B-D)=7	{A,B,C,D,E,F,G}

New MST edges: (A-C)=6, (B-C)=5, (E-F)=5, (D-E)=5, (F-G)=6, (B-D)=7

New total weight: 34 (was 32 → +2, a 6% increase)

Graph still fully connected: True

Scenario 1: Edge A-G Fails (Total Weight = 34)

